

BIOS5801: STATISTICAL COMPUTING

Effective Term

Semester A 2025/26

Part I Course Overview

Course Title

Statistical Computing

Subject Code

BIOS - Biostatistics

Course Number

5801

Academic Unit

Biostatistics (BIOS)

College/School

College of Computing (CC)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

Nil

Precursors

Nil

Equivalent Courses

Nil

Exclusive Courses

Nil

Part II Course Details

Abstract

Contemporary biostatistics and data analysis depend on a mastery of tools for computation, visualization, dissemination, and reproducibility, in addition to proficiency in traditional statistical techniques. The goal of this course is to provide

training in the elements of a complete pipeline for data analysis using R. It is targeted to students with some data analysis experience.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Understand the importance of visualization, dissemination and reproducibility in data analysis	40	x	x	
2	Ability to provide a complete pipeline for data analysis using R	40	x	x	x
3	Appreciate the relevance of statistical computing for applications in public health	20	x	x	x

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Teaching	Learning through teaching based on lectures	1, 2, 3	3 hours/ week
2	Assignments	Learning through assignments allows students to develop hands-on skills involving data analysis using R	1, 2, 3	

Assessment Tasks / Activities (ATs)

ATs		CILO No.	Weighting (%)	Remarks ("- " for nil entry)	Allow Use of GenAI?
1	Assignments	1, 2, 3	30	-	Yes
2	Project	1, 2, 3	30	-	No

Continuous Assessment (%)

60

Examination (%)

40

Examination Duration (Hours)

2

Minimum Continuous Assessment Passing Requirement (%)

40

Minimum Examination Passing Requirement (%)

35

Additional Information for ATs

To pass the course, students are required to obtain a minimum of 40% in continuous assessment and a minimum of 35% in the examination.

Assessment Rubrics (AR)

Assessment Task

Assignments (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Problem solving skills

Excellent

(A+, A, A-) Consistently demonstrates a thorough understanding of data analysis and strong ability to solve problems using R

Good

(B+, B, B-) Adequately demonstrates an understanding of data analysis using R and ability to solve problems using R

Fair

(C+, C, C-) Demonstrates some understanding of data analysis by using R to solve simple problems

Marginal

(D) Demonstrates some understanding of data analysis but cannot apply R to solve simple problems

Failure

(F) Demonstrates little understanding of data analysis using R and is unable to apply them to problems

Assessment Task

Project (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Problem solving based on comprehensive understanding

Excellent

(A+, A, A-) Demonstrates a comprehensive understanding of data analysis and strong ability in applying R to solve relevant problems in public health

Good

(B+, B, B-) Adequately demonstrates an understanding of data analysis and ability in applying R to solve relevant problems in public health

Fair

(C+, C, C-) Demonstrates some understanding of data analysis and little ability to solve simple problems with limited success

Marginal

(D) Demonstrates some understanding of data analysis but cannot apply R to solve simple problems

Failure

(F) Inappropriately or unable to apply data analysis using R to solve problems

Assessment Task

Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Problem solving based on comprehensive understanding

Excellent

(A+, A, A-) Consistently demonstrates a comprehensive understanding of data analysis and strong ability in applying relevant functions in R to solve complex problems in public health

Good

(B+, B, B-) Adequately demonstrates an understanding of data analysis and ability in applying relevant functions in R to solve complex problems in public health

Fair

(C+, C, C-) Demonstrates some understanding of data analysis and ability in using R to solve simple problems

Marginal

(D) Demonstrates some understanding of data analysis using R and limited ability in using R to solve simple problems

Failure

(F) Demonstrates little understanding of data analysis using R and is unable to apply them to problems

Assessment Task

Assignment (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Problem solving skills

Excellent

(A+, A, A-) Consistently demonstrates a thorough understanding of data analysis and strong ability to solve problems using R

Good

(B+, B) Adequately demonstrates an understanding of data analysis using R and ability to solve problems using R

Marginal

(B-, C+, C) Demonstrates some understanding of data analysis by using R to solve simple problems

Failure

(F) Demonstrates little understanding of data analysis using R and is unable to apply them to problems

Assessment Task

Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Problem solving based on comprehensive understanding

Excellent

(A+, A, A-) Demonstrates a comprehensive understanding of data analysis and strong ability in applying R to solve relevant problems in public health

Good

(B+, B) Adequately demonstrates an understanding of data analysis and ability in applying R to solve relevant problems in public health

Marginal

(B-, C+, C) Demonstrates some understanding of data analysis and little ability to solve simple problems with limited success

Failure

(F) Inappropriately or unable to apply data analysis using R to solve problems

Assessment Task

Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Problem solving based on comprehensive understanding

Excellent

(A+, A, A-) Consistently demonstrates a comprehensive understanding of data analysis and strong ability in applying relevant functions in R to solve complex problems in public health

Good

(B+, B) Adequately demonstrates an understanding of data analysis and ability in applying relevant functions in R to solve complex problems in public health

Marginal

(B-, C+, C) Demonstrates some understanding of data analysis and ability in using R to solve simple problems

Failure

(F) Demonstrates little understanding of data analysis using R and is unable to apply them to problems

Part III Other Information

Keyword Syllabus

Visualization, dissemination and reproducibility for data analysis using R.

Reading List

Compulsory Readings

Title	
1	R for Data Science: Import, Tidy, Transform, Visualize, and Model Data by Hadley Wickham and Garrett Grolemund

Additional Readings

Title	
1	Nil